CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

M & A DAIRY #3 (FORMERLY DEN-K HOLSTEINS)

13315 BAKER RD

LOS BANOS CA 93635

NOTES:

HOME 2, AND NEMETHI FIELDS 2, 5, 8/9 AND WW1 WERE SUMMER FALLOWED

PREPARED BY:

CAS
CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: M and A Dairy #3

Physical address of dairy:

Number and Street	City	County	Zip Code
13235 Baker RD	Los Banos	Merced	93635

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1981

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X073-X200-X004-XXXX

B. OPERATORS

Areias, Dennis & Kirsten

Operator name: Areias, Dennis & Kirsten

Telephone no.: (209) 826-6242 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
13315 Baker RD	Los Banos	CA	93635

Gonclaves, Anthony

Operator name: Gonclaves, Anthony

Telephone no.: (209) 769-8316 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
14720 Carlucci	Dos Palos	CA	93620

This operator is responsible for paying permit fees.

C. OWNERS

Areias, Dennis & Kirsten

Legal owner name: Areias, Dennis & Kirsten

Telephone no.: (209) 826-6242 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
13315 Baker RD	Los Banos	CA	93635

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	374	0	0	0	0	0
Maximum number	374	0	0	0	0	0
Average number	374	0	0	0	0	0
Avg live weight (lbs)	1,300	0	0	0		

Predominant milk cow breed: Holstein

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 9,505.34 *tons per reporting period*

Total nitrogen from manure: 124,284.38 *lbs per reporting period*

After ammonia losses (30% loss applied): 86,999.07 *lbs per reporting period*

Total phosphorus from manure: 21,023.87 *lbs per reporting period*

Total potassium from manure: 71,133.01 *lbs per reporting period*

Total salt from manure: 176,097.90 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 22,900,500 *gallons*

Total nitrogen generated: 58,772.48 *lbs*

Total phosphorus generated: 7,666.43 *lbs*

Total potassium generated: 195,617.73 *lbs*

Total salt generated: 1,148,490.64 *lbs*

	22,900,500	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	22,900,500	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
HOME 1	37	37	2	process wastewater	X073-X200-X004-XXXX
HOME 2	37	37	1	process wastewater	X073-X200-X006-XXXX
NEMETHI 1	20	20	2	manure	X073-X160-X007-XXXX
NEMETHI 2	18	18	1	manure	X073-X160-X008-XXXX
NEMETHI 3	12	12	2	manure	X073-X160-X007-XXXX
NEMETHI 4	35	35	2	manure	X073-X160-X007-XXXX X073-X160-X008-XXXX
NEMETHI 5	19	19	1	manure	X073-X160-X008-XXXX
NEMETHI 6	13	13	2	manure	X073-X160-X008-XXXX
NEMETHI 8/9	11	11	1	manure	X073-X160-X007-XXXX X073-X160-X008-XXXX
SANDERS	27	27	2	process wastewater	X073-X200-X002-XXXX X073-X200-X022-XXXX
WW1	18	18	1	process wastewater	X073-X200-X022-XXXX
WW2	20	20	2	process wastewater	X073-X200-X022-XXXX
Totals for areas that were used for application	267	267	19		
Totals for areas that were not used for application					
Land application area totals	267	267	19		

B. CROPS AND HARVESTS

HOME 1

Field name: HOME 1

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	544.00 *ton*	Dry-weight		41.3	12,900.00	2,400.00	14,900.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.70	222.67	41.43	257.19	1,467.18

HOME 1

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	748.00 *ton*	Dry-weight		51.6	11,800.00	2,000.00	10,900.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.22	230.92	39.14	213.31	1,095.88

HOME 2

Field name: HOME 2

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	592.00 *ton*	Dry-weight		41.3	12,900.00	2,400.00	4,900.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.00	242.31	45.08	92.04	1,596.64

NEMETHI 1

Field name: NEMETHI 1

NEMETHI 1

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	290.00 *ton*	Dry-weight		74.5	17,200.00	3,300.00	23,000.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.50	127.19	24.40	170.09	1,020.51

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	249.00 *ton*	Dry-weight		50.9	11,400.00	2,200.00	9,450.00		4.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	12.45	139.38	26.90	115.53	495.15

NEMETHI 2

Field name: NEMETHI 2

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	266.00 *ton*	Dry-weight		74.5	17,200.00	3,300.00	23,000.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.78	129.63	24.87	173.34	1,040.06

NEMETHI 3

Field name: NEMETHI 3

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	166.00 *ton*	Dry-weight		74.5	17,200.00	3,300.00	23,000.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.83	121.35	23.28	162.27	973.59

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	297.00 *ton*	Dry-weight		50.9	11,400.00	2,200.00	9,450.00		4.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.75	277.07	53.47	229.68	984.33

NEMETHI 4

Field name: NEMETHI 4

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	500.00 *ton*	Dry-weight		74.5	17,200.00	3,300.00	23,000.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.29	125.31	24.04	167.57	1,005.43

NEMETHI 4

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	999.00 *ton*	Dry-weight		50.9	11,400.00	2,200.00	9,450.00		4.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.54	319.53	61.66	264.87	1,135.18

NEMETHI 5

Field name: NEMETHI 5

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	285.00 *ton*	Dry-weight		74.6	17,700.00	3,500.00	23,100.00		14.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.00	134.87	26.67	176.02	1,104.90

NEMETHI 6

Field name: NEMETHI 6

NEMETHI 6

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 13 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	185.00 *ton*	Dry-weight		74.6	17,700.00	3,500.00	23,100.00		14.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.23	127.96	25.30	167.00	1,048.24

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	369.00 *ton*	Dry-weight		50.9	11,400.00	2,200.00	9,450.00		4.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.38	317.76	61.32	263.41	1,128.88

NEMETHI 8/9

Field name: NEMETHI 8/9

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	228.00 *ton*	Dry-weight		74.6	17,700.00	3,500.00	23,100.00		14.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	20.73	186.37	36.85	243.23	1,526.77

SANDERS

Field name: SANDERS

10/27/2022: Ryegrass Silage

Crop: Ryegrass Silage Acres planted: 27 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	390.00 *ton*	Dry-weight		77.2	11,500.00	3,300.00	32,500.00		13.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.44	75.75	21.74	214.07	915.55

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	832.00 *ton*	Dry-weight		58.4	11,600.00	2,000.00	10,300.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.81	297.40	51.28	264.07	1,384.45

WW1

Field name: WW1

10/27/2022: Ryegrass Silage

Crop: Ryegrass Silage Acres planted: 18 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	252.00 *ton*	Dry-weight		74.9	13,800.00	3,500.00	34,200.00		16.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	96.99	24.60	240.36	1,166.65

WW2

Field name: WW2

10/27/2022: Ryegrass Silage

Crop: Ryegrass Silage Acres planted: 20 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	305.00 *ton*	Dry-weight		74.9	13,800.00	3,500.00	34,200.00		16.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.25	105.65	26.79	261.82	1,270.81

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	544.00 *ton*	Dry-weight		58.4	11,600.00	2,000.00	10,300.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.20	262.51	45.26	233.09	1,222.04

NUTRIENT BUDGET

A. LAND APPLICATIONS

HOME 1 - 10/27/2022: Oats, silage-soft dough

Field name: HOME 1

Crop: Oats, silage-soft dough — Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.20	5.96	343.86	1,786.28	990,000.00 *gal*
Application event totals		94.20	5.96	343.86	1,786.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.18	10.74	523.46	2,591.73	2,520,000.00 *gal*
Application event totals		99.18	10.74	523.46	2,591.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	7,892,157.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.72	11.13	542.16	2,684.29	2,610,000.00 *gal*
Canal	Surface water	0.09	0.00	0.00	0.00	3,938,040.00 *gal*
Application event totals		102.81	11.13	542.16	2,684.29	

HOME 1 - 05/21/2023: Corn, silage

Field name: HOME 1

Crop: Corn, silage — Plant date: 05/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

HOME 1 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	106.18	18.15	224.20	1,712.50	1,057,500.00 *gal*
Canal	Surface water	0.12	0.00	0.00	0.00	5,150,555.00 *gal*
Application event totals		106.30	18.15	224.20	1,712.50	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	13,000,099.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	106.18	18.15	224.20	1,712.50	1,057,500.00 *gal*
Canal	Surface water	0.15	0.00	0.00	0.00	6,665,069.00 *gal*
Application event totals		106.33	18.15	224.20	1,712.50	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	10,378,335.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	5,402,291.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

HOME 1 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.93	17.76	219.43	1,676.06	1,035,000.00 *gal*
Canal	Surface water	0.12	0.00	0.00	0.00	5,524,815.00 *gal*
Application event totals		104.05	17.76	219.43	1,676.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	3,198,577.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

HOME 2 - 10/27/2022: Oats, silage-soft dough

Field name: HOME 2

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	111.33	7.05	406.38	2,111.06	1,170,000.00 *gal*
Application event totals		111.33	7.05	406.38	2,111.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.01	9.21	448.68	2,221.48	2,160,000.00 *gal*
Application event totals		85.01	9.21	448.68	2,221.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	7,892,157.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

HOME 2 - 10/27/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.01	9.21	448.68	2,221.48	2,160,000.00 *gal*
Canal	Surface water	0.09	0.00	0.00	0.00	3,938,040.00 *gal*
Application event totals		85.10	9.21	448.68	2,221.48	

NEMETHI 1 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 1

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	169.95	39.94	56.08	0.00	427.00 *ton*
Application event totals		169.95	39.94	56.08	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	4,266,031.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	2,128,670.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

NEMETHI 1 - 05/25/2023: Corn, silage

Field name: NEMETHI 1

Crop: Corn, silage

Plant date: 05/25/2023

NEMETHI 1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	185.82	60.57	130.38	4,003.86	549.00 *ton*
Application event totals		185.82	60.57	130.38	4,003.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	2,784,084.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	7,027,080.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	3,602,740.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	5,609,911.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	2,920,157.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

NEMETHI 1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	2,986,387.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	1,728,961.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

NEMETHI 2 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 2

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	166.72	39.18	55.02	0.00	377.00 *ton*
Application event totals		166.72	39.18	55.02	0.00	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	3,839,428.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	1,915,803.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

NEMETHI 3 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 3

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	161.85	38.04	53.41	0.00	244.00 *ton*
Application event totals		161.85	38.04	53.41	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	2,559,619.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	1,277,202.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

NEMETHI 3 - 05/25/2023: Corn, silage

Field name: NEMETHI 3

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	330.01	107.57	231.55	7,110.68	585.00 *ton*
Application event totals		330.01	107.57	231.55	7,110.68	

NEMETHI 3 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	1,670,450.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	4,216,248.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	2,161,644.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	3,365,947.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	1,752,094.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	1,791,832.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

NEMETHI 3 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	1,037,376.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

NEMETHI 4 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 4

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	168.07	39.50	55.46	0.00	739.00 *ton*
Application event totals		168.07	39.50	55.46	0.00	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	7,465,554.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	3,725,173.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

NEMETHI 4 - 05/25/2023: Corn, silage

Field name: NEMETHI 4

Crop: Corn, silage

Plant date: 05/25/2023

NEMETHI 4 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	329.96	107.56	231.52	7,109.63	1,706.00 *ton*
Application event totals		329.96	107.56	231.52	7,109.63	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	4,872,147.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	12,297,391.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	6,304,795.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	9,817,344.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	5,110,275.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

NEMETHI 4 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	5,226,177.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	3,025,681.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

NEMETHI 5 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 5

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	174.70	41.05	57.65	0.00	417.00 *ton*
Application event totals		174.70	41.05	57.65	0.00	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	4,052,729.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	2,022,237.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

NEMETHI 6 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 6

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	171.45	40.29	56.58	0.00	280.00 *ton*
Application event totals		171.45	40.29	56.58	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	2,772,920.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	1,383,636.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

NEMETHI 6 - 05/25/2023: Corn, silage

Field name: NEMETHI 6

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	330.14	107.61	231.64	7,113.48	634.00 *ton*
Application event totals		330.14	107.61	231.64	7,113.48	

NEMETHI 6 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	1,809,654.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.29	0.00	0.00	0.00	4,567,602.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.15	0.00	0.00	0.00	2,341,781.00 *gal*
Application event totals		0.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	3,646,442.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	1,898,102.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	1,941,151.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

NEMETHI 6 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	1,123,824.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

NEMETHI 8/9 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 8/9

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	242.42	56.97	80.00	0.00	335.00 *ton*
Application event totals		242.42	56.97	80.00	0.00	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	2,346,317.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.09	0.00	0.00	0.00	1,170,769.00 *gal*
Application event totals		0.09	0.00	0.00	0.00	

SANDERS - 10/27/2022: Ryegrass Silage

Field name: SANDERS

Crop: Ryegrass Silage

Plant date: 10/27/2022

SANDERS - 10/27/2022: Ryegrass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.65	4.34	250.60	1,301.82	526,500.00 *gal*
Application event totals		68.65	4.34	250.60	1,301.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	5,759,142.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.12	3.15	153.71	761.06	540,000.00 *gal*
Canal	Surface water	0.09	0.00	0.00	0.00	2,873,705.00 *gal*
Application event totals		29.21	3.15	153.71	761.06	

SANDERS - 05/21/2023: Corn, silage

Field name: SANDERS

Crop: Corn, silage

Plant date: 05/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.93	17.25	213.10	1,627.75	733,500.00 *gal*
Canal	Surface water	0.12	0.00	0.00	0.00	3,758,513.00 *gal*
Application event totals		101.05	17.25	213.10	1,627.75	

SANDERS - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.93	17.25	213.10	1,627.75	733,500.00 *gal*
Canal	Surface water	0.29	0.00	0.00	0.00	9,486,558.00 *gal*
Application event totals		101.22	17.25	213.10	1,627.75	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.93	17.25	213.10	1,627.75	733,500.00 *gal*
Canal	Surface water	0.15	0.00	0.00	0.00	4,863,699.00 *gal*
Application event totals		101.08	17.25	213.10	1,627.75	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	7,573,380.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	3,942,212.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.93	17.25	213.10	1,627.75	733,500.00 *gal*
Canal	Surface water	0.12	0.00	0.00	0.00	4,031,622.00 *gal*
Application event totals		101.05	17.25	213.10	1,627.75	

SANDERS - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	2,334,097.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

WW1 - 10/27/2022: Ryegrass Silage

Field name: WW1

Crop: Ryegrass Silage Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.38	5.40	311.64	1,618.93	436,500.00 *gal*
Application event totals		85.38	5.40	311.64	1,618.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.18	0.00	0.00	0.00	3,839,428.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.41	3.94	192.14	951.33	450,000.00 *gal*
Canal	Surface water	0.09	0.00	0.00	0.00	1,915,803.00 *gal*
Application event totals		36.49	3.94	192.14	951.33	

WW2 - 10/27/2022: Ryegrass Silage

Field name: WW2

Crop: Ryegrass Silage Plant date: 10/27/2022

WW2 - 10/27/2022: Ryegrass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.26	4.76	274.70	1,427.00	427,500.00 *gal*
Application event totals		75.26	4.76	274.70	1,427.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	32.76	3.55	172.93	856.20	450,000.00 *gal*
Canal	Surface water	0.18	0.00	0.00	0.00	4,266,031.00 *gal*
Application event totals		32.94	3.55	172.93	856.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	32.76	3.55	172.93	856.20	450,000.00 *gal*
Canal	Surface water	0.09	0.00	0.00	0.00	2,128,670.00 *gal*
Application event totals		32.85	3.55	172.93	856.20	

WW2 - 05/21/2023: Corn, silage

Field name: WW2

Crop: Corn, silage

Plant date: 05/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.44	15.29	188.85	1,442.50	481,500.00 *gal*
Canal	Surface water	0.12	0.00	0.00	0.00	2,784,084.00 *gal*
Application event totals		89.56	15.29	188.85	1,442.50	

WW2 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.44	15.29	188.85	1,442.50	481,500.00 *gal*
Canal	Surface water	0.29	0.00	0.00	0.00	7,027,080.00 *gal*
Application event totals		89.74	15.29	188.85	1,442.50	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.44	15.29	188.85	1,442.50	481,500.00 *gal*
Canal	Surface water	0.15	0.00	0.00	0.00	3,602,740.00 *gal*
Application event totals		89.59	15.29	188.85	1,442.50	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.23	0.00	0.00	0.00	5,609,911.00 *gal*
Application event totals		0.23	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.12	0.00	0.00	0.00	2,920,157.00 *gal*
Application event totals		0.12	0.00	0.00	0.00	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.44	15.29	188.85	1,442.50	481,500.00 *gal*
Canal	Surface water	0.12	0.00	0.00	0.00	2,986,387.00 *gal*
Application event totals		89.57	15.29	188.85	1,442.50	

WW2 - 05/21/2023: Corn, silage

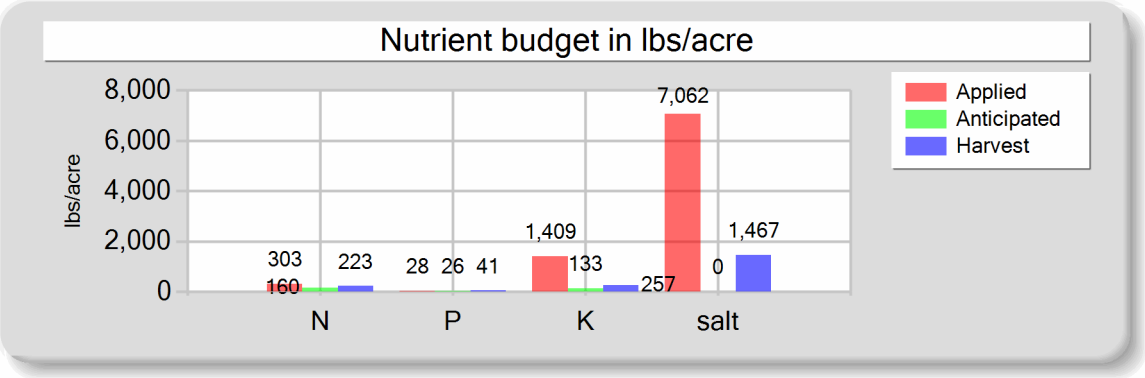
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.07	0.00	0.00	0.00	1,728,961.00 *gal*
Application event totals		0.07	0.00	0.00	0.00	

B. NUTRIENT BUDGET

HOME 1 - 10/27/2022: Oats, silage-soft dough

Field name: HOME 1 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	296.10	27.83	1,409.48	7,062.31
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	303.37	27.83	1,409.48	7,062.31
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	222.67	41.43	257.19	1,467.18
Nutrient balance	80.70	-13.60	1,152.29	5,595.12
Applied to removed ratio	1.36	0.67	5.48	4.81

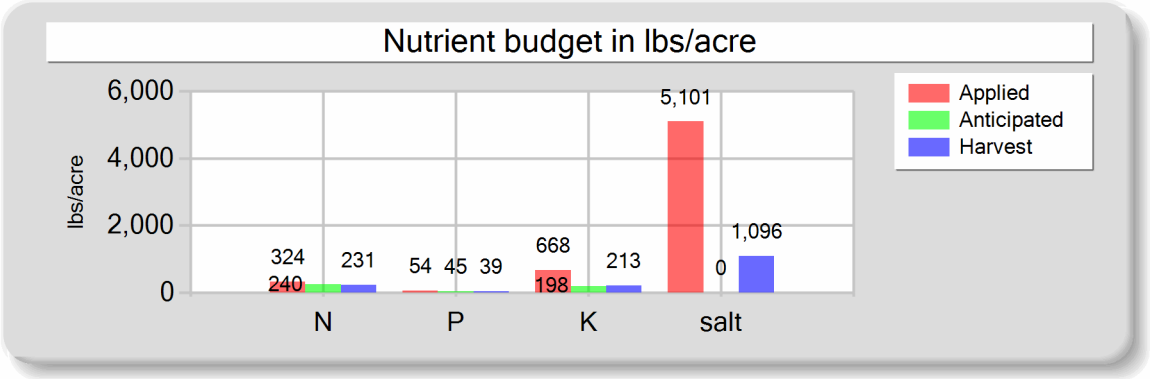
Fresh water applied	
11,830,197.00	*gallons*
435.67	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
6,120,000.00	*gallons*
225.38	*acre-inches*
6.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

HOME 1 - 05/21/2023: Corn, silage

Field name: HOME 1 Crop: Corn, silage Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	316.29	54.07	667.83	5,101.05
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	324.41	54.07	667.83	5,101.05
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	230.92	39.14	213.31	1,095.88
Nutrient balance	93.49	14.93	454.52	4,005.17
Applied to removed ratio	1.40	1.38	3.13	4.65

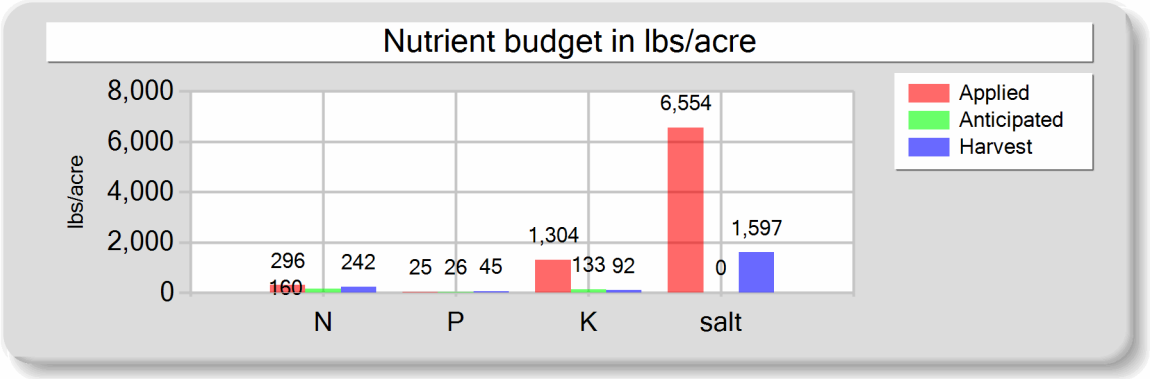
Fresh water applied	
49,319,741.00	*gallons*
1,816.28	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
3,150,000.00	*gallons*
116.00	*acre-inches*
3.14	*inches/acre*

Total harvests for the crop	
1	*harvests*

HOME 2 - 10/27/2022: Oats, silage-soft dough

Field name: HOME 2 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	281.35	25.46	1,303.74	6,554.03
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	295.62	25.46	1,303.74	6,554.03
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	242.31	45.08	92.04	1,596.64
Nutrient balance	53.31	-19.62	1,211.70	4,957.39
Applied to removed ratio	1.22	0.56	14.16	4.10

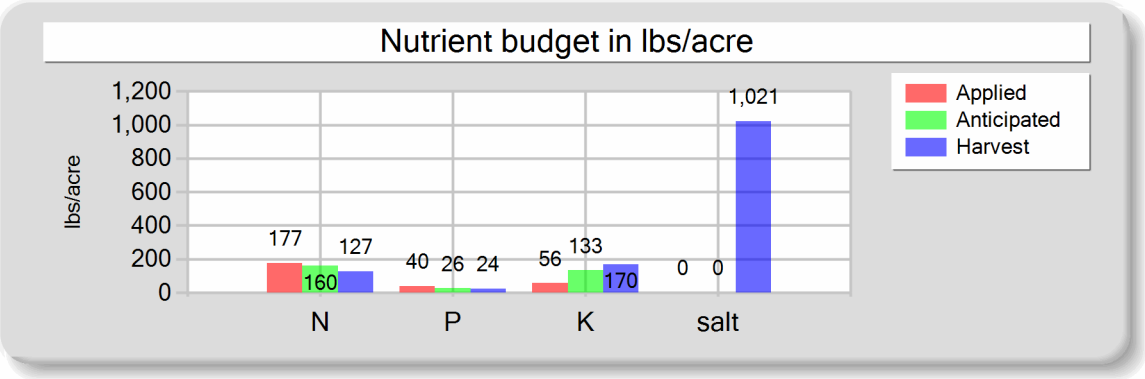
Fresh water applied	
11,830,197.00	*gallons*
435.67	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
5,490,000.00	*gallons*
202.18	*acre-inches*
5.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEMETHI 1 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 1 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	169.95	39.94	56.08	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	177.21	39.94	56.08	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	127.19	24.40	170.09	1,020.51
Nutrient balance	50.02	15.53	-114.00	-1,020.51
Applied to removed ratio	1.39	1.64	0.33	0.00

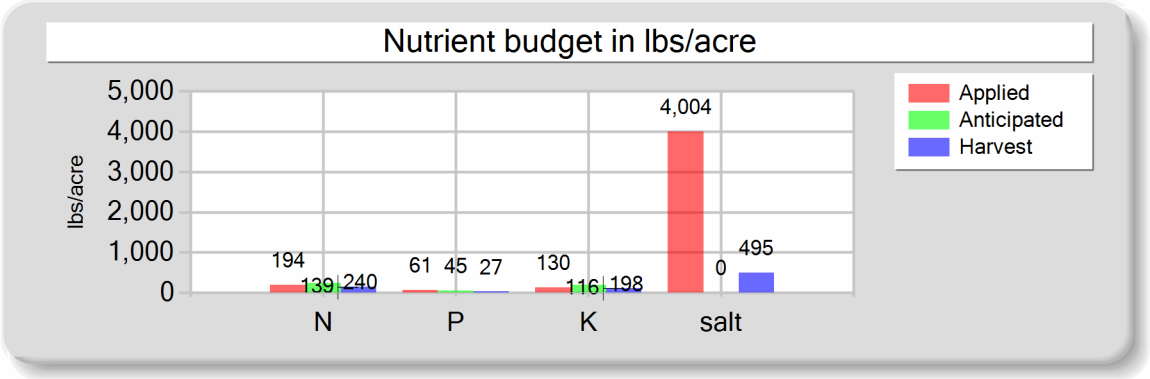
Fresh water applied	
6,394,701.00	*gallons*
235.50	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEMETHI 1 - 05/25/2023: Corn, silage

Field name: NEMETHI 1 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	185.82	60.57	130.38	4,003.86
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	193.93	60.57	130.38	4,003.86
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	139.38	26.90	115.53	495.15
Nutrient balance	54.56	33.67	14.85	3,508.71
Applied to removed ratio	1.39	2.25	1.13	8.09

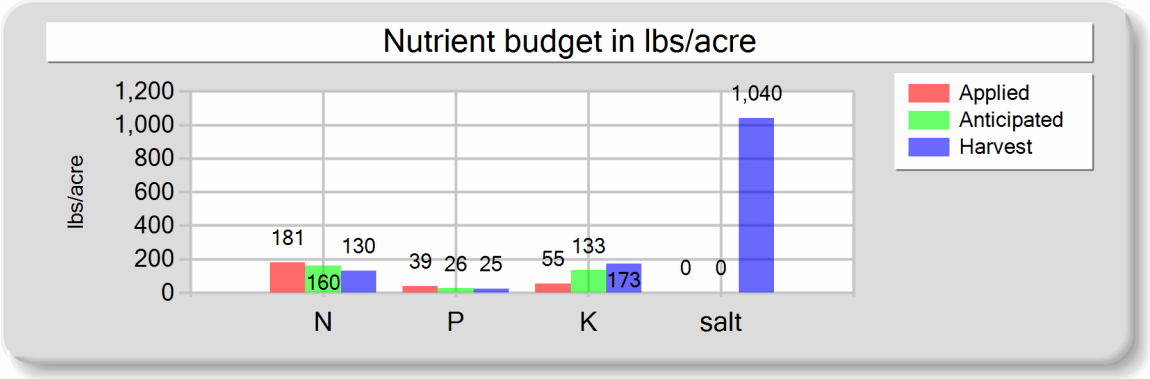
Fresh water applied	
26,659,320.00	*gallons*
981.77	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEMETHI 2 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 2 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	166.72	39.18	55.02	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	180.98	39.18	55.02	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	129.63	24.87	173.34	1,040.06
Nutrient balance	51.35	14.31	-118.33	-1,040.06
Applied to removed ratio	1.40	1.58	0.32	0.00

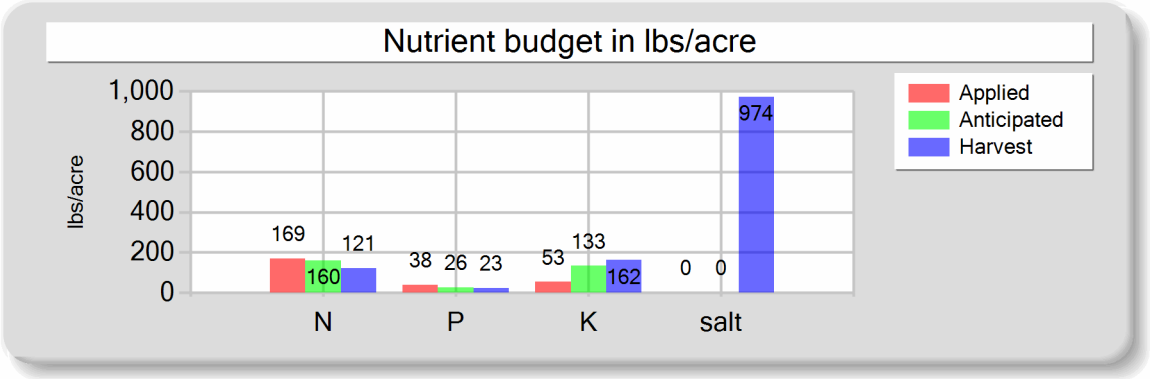
Fresh water applied	
5,755,231.00	*gallons*
211.95	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEMETHI 3 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 3 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	161.85	38.04	53.41	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	169.12	38.04	53.41	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	121.35	23.28	162.27	973.59
Nutrient balance	47.77	14.75	-108.85	-973.59
Applied to removed ratio	1.39	1.63	0.33	0.00

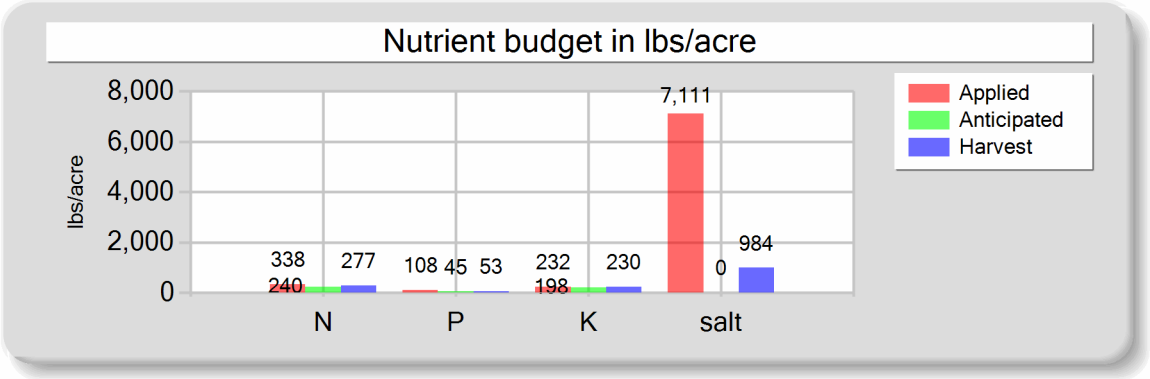
Fresh water applied	
3,836,821.00	*gallons*
141.30	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEMETHI 3 - 05/25/2023: Corn, silage

Field name: NEMETHI 3 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	330.01	107.57	231.55	7,110.68
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	338.12	107.57	231.55	7,110.68
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	277.07	53.47	229.68	984.33
Nutrient balance	61.05	54.10	1.88	6,126.34
Applied to removed ratio	1.22	2.01	1.01	7.22

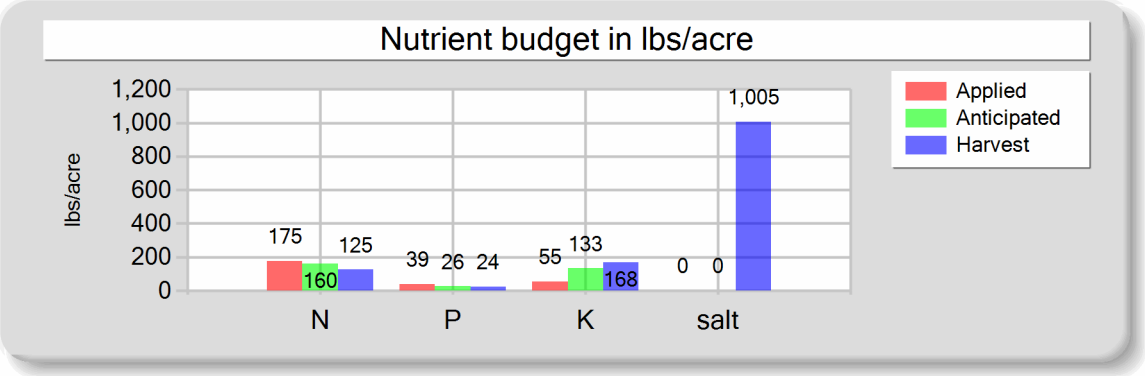
Fresh water applied	
15,995,591.00	*gallons*
589.06	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEMETHI 4 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 4 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	168.07	39.50	55.46	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	175.34	39.50	55.46	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	125.31	24.04	167.57	1,005.43
Nutrient balance	50.02	15.45	-112.11	-1,005.43
Applied to removed ratio	1.40	1.64	0.33	0.00

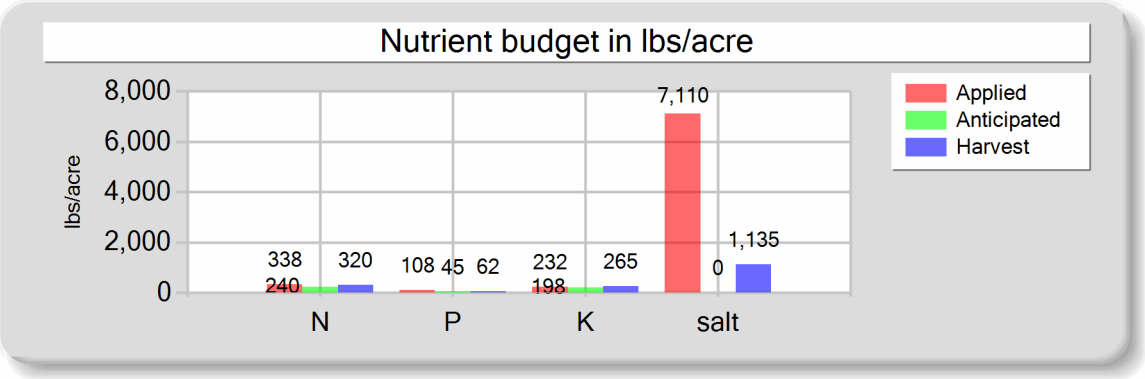
Fresh water applied	
11,190,727.00	*gallons*
412.12	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEMETHI 4 - 05/25/2023: Corn, silage

Field name: NEMETHI 4 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	329.96	107.56	231.52	7,109.63
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	338.07	107.56	231.52	7,109.63
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	319.53	61.66	264.87	1,135.18
Nutrient balance	18.54	45.89	-33.36	5,974.46
Applied to removed ratio	1.06	1.74	0.87	6.26

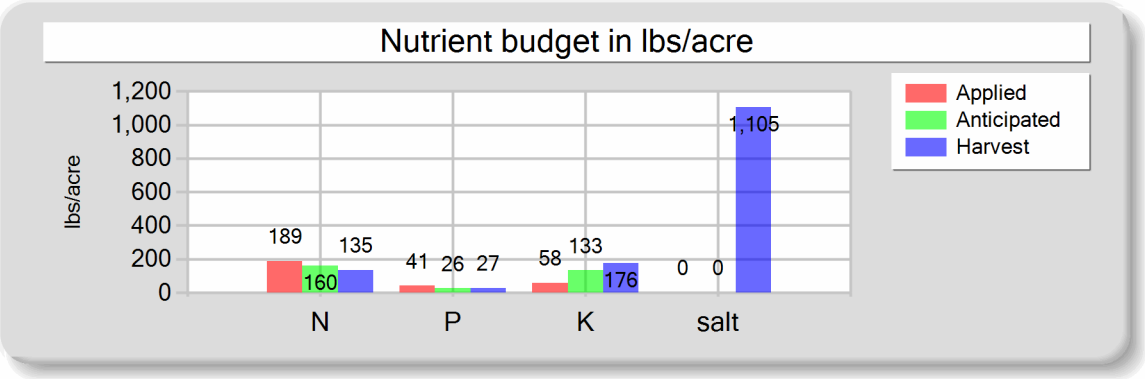
Fresh water applied	
46,653,810.00	*gallons*
1,718.10	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEMETHI 5 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 5 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	174.70	41.05	57.65	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	188.97	41.05	57.65	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	134.87	26.67	176.02	1,104.90
Nutrient balance	54.09	14.38	-118.37	-1,104.90
Applied to removed ratio	1.40	1.54	0.33	0.00

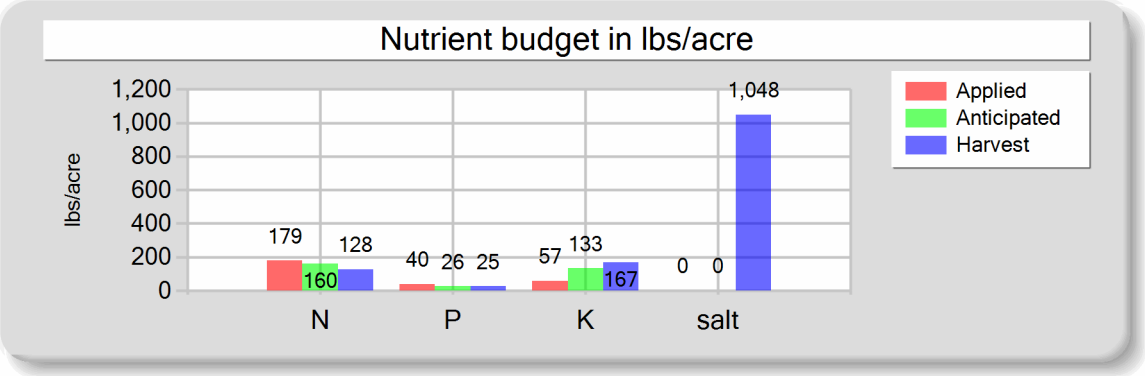
Fresh water applied	
6,074,966.00	*gallons*
223.72	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEMETHI 6 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 6 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	171.45	40.29	56.58	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	178.71	40.29	56.58	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	127.96	25.30	167.00	1,048.24
Nutrient balance	50.76	14.99	-110.42	-1,048.24
Applied to removed ratio	1.40	1.59	0.34	0.00

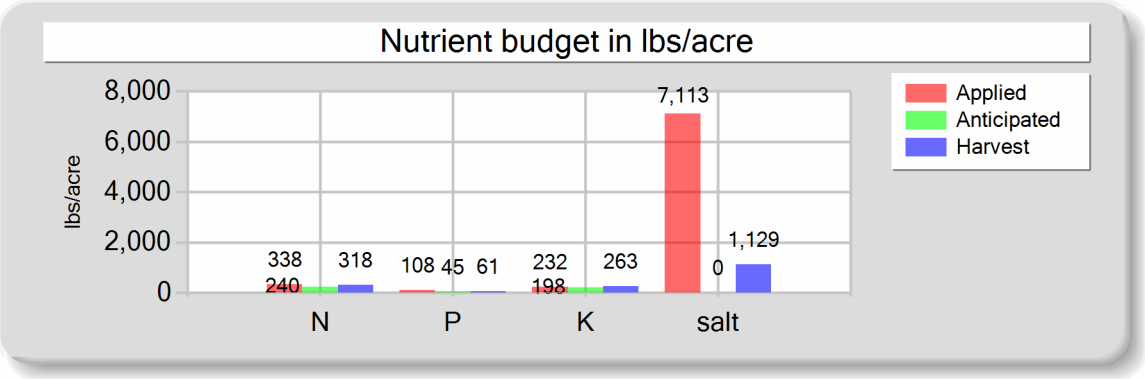
Fresh water applied	
4,156,556.00	*gallons*
153.07	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEMETHI 6 - 05/25/2023: Corn, silage

Field name: NEMETHI 6 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	330.14	107.61	231.64	7,113.48
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	338.25	107.61	231.64	7,113.48
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	317.76	61.32	263.41	1,128.88
Nutrient balance	20.49	46.29	-31.76	5,984.60
Applied to removed ratio	1.06	1.75	0.88	6.30

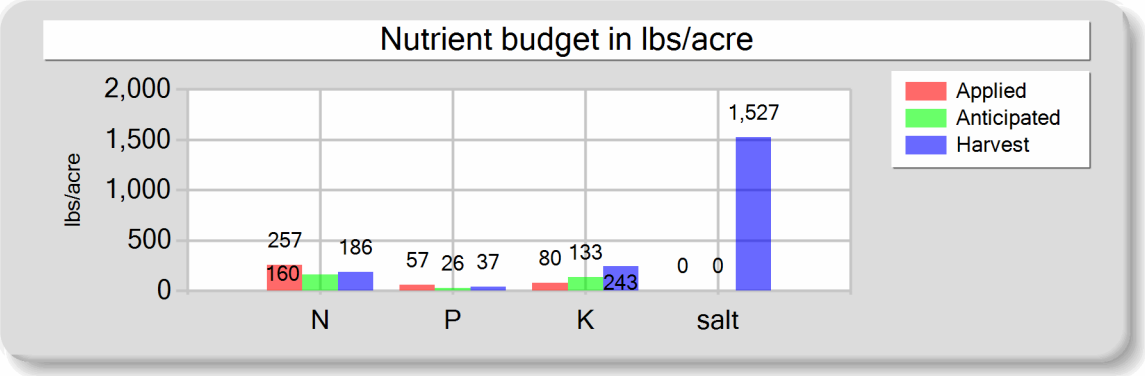
Fresh water applied	
17,328,556.00	*gallons*
638.15	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NEMETHI 8/9 - 10/20/2022: Oats, silage-soft dough

Field name: NEMETHI 8/9 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	242.42	56.97	80.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	256.69	56.97	80.00	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	186.37	36.85	243.23	1,526.77
Nutrient balance	70.31	20.12	-163.23	-1,526.77
Applied to removed ratio	1.38	1.55	0.33	0.00

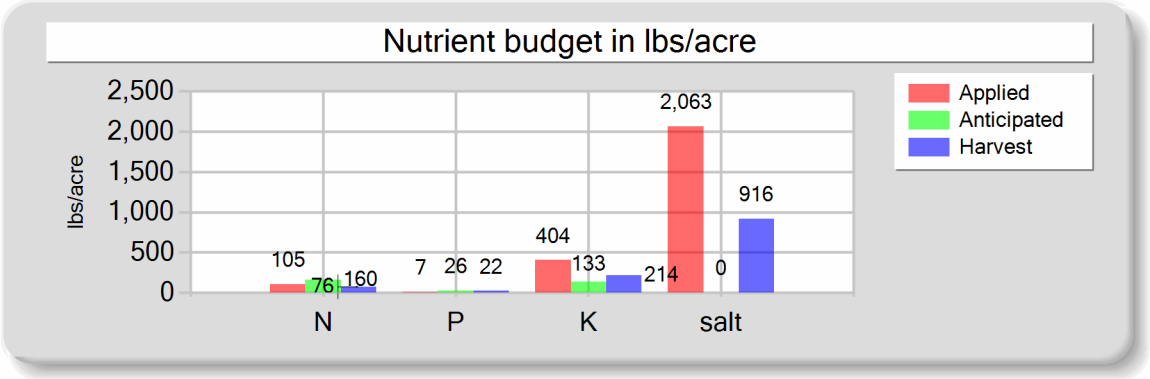
Fresh water applied	
3,517,086.00	*gallons*
129.52	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

SANDERS - 10/27/2022: Ryegrass Silage

Field name: SANDERS | Crop: Ryegrass Silage | Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	97.78	7.50	404.32	2,062.88
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	105.05	7.50	404.32	2,062.88
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	75.75	21.74	214.07	915.55
Nutrient balance	29.30	-14.24	190.25	1,147.34
Applied to removed ratio	1.39	0.35	1.89	2.25

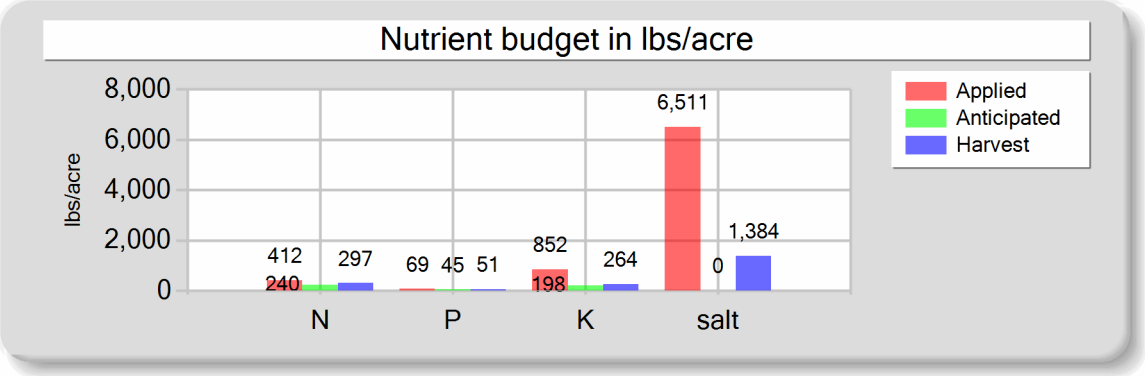
Fresh water applied	
8,632,847.00	*gallons*
317.92	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
1,066,500.00	*gallons*
39.28	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

SANDERS - 05/21/2023: Corn, silage

Field name: SANDERS Crop: Corn, silage Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	403.72	69.01	852.41	6,510.99
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	411.83	69.01	852.41	6,510.99
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	297.40	51.28	264.07	1,384.45
Nutrient balance	114.43	17.73	588.34	5,126.54
Applied to removed ratio	1.38	1.35	3.23	4.70

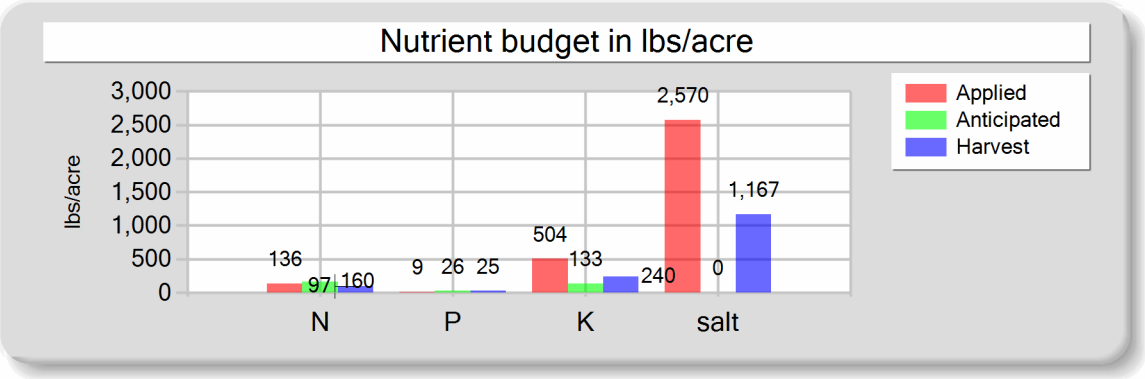
Fresh water applied	
35,990,081.00	*gallons*
1,325.39	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
2,934,000.00	*gallons*
108.05	*acre-inches*
4.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

WW1 - 10/27/2022: Ryegrass Silage

Field name: WW1 Crop: Ryegrass Silage Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	121.78	9.35	503.79	2,570.26
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	136.05	9.35	503.79	2,570.26
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	96.99	24.60	240.36	1,166.65
Nutrient balance	39.06	-15.25	263.43	1,403.61
Applied to removed ratio	1.40	0.38	2.10	2.20

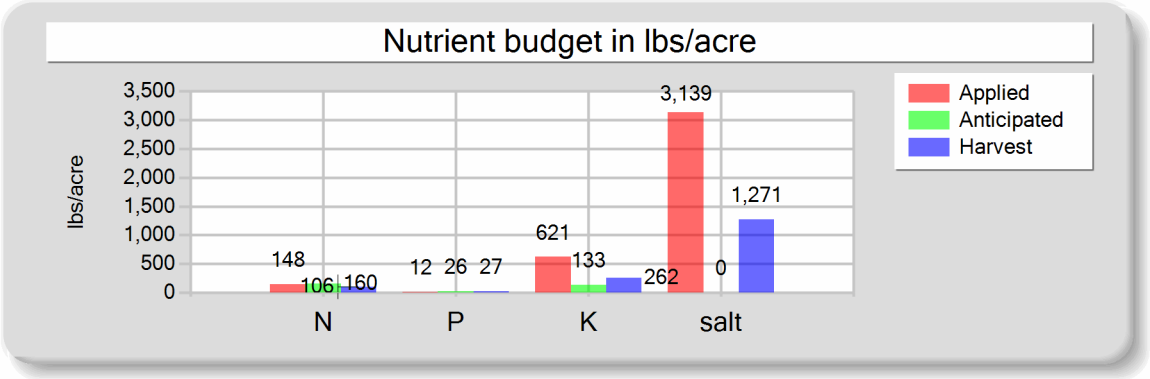
Fresh water applied	
5,755,231.00	*gallons*
211.95	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
886,500.00	*gallons*
32.65	*acre-inches*
1.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

WW2 - 10/27/2022: Ryegrass Silage

Field name: WW2 Crop: Ryegrass Silage Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	140.79	11.86	620.56	3,139.39
Fresh water	0.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	148.05	11.86	620.56	3,139.39
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	105.65	26.79	261.82	1,270.81
Nutrient balance	42.41	-14.93	358.74	1,868.58
Applied to removed ratio	1.40	0.44	2.37	2.47

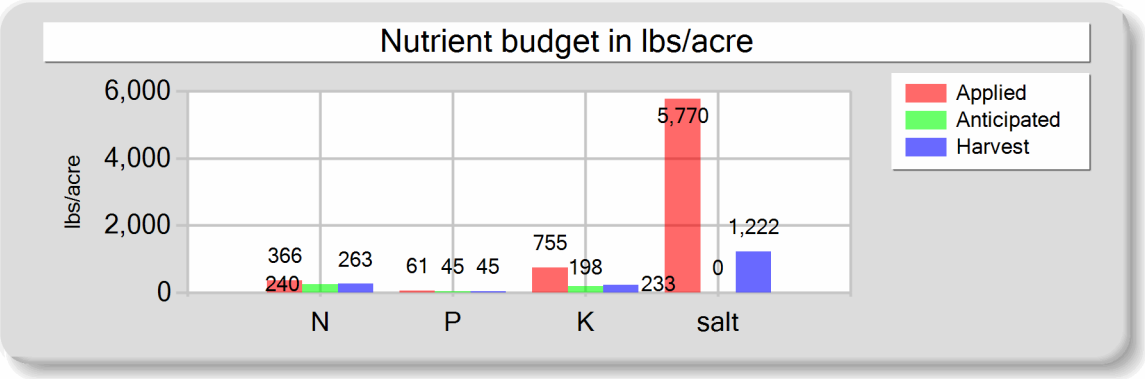
Fresh water applied	
6,394,701.00	*gallons*
235.50	*acre-inches*
11.77	*inches/acre*

Process wastewater applied	
1,327,500.00	*gallons*
48.89	*acre-inches*
2.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

WW2 - 05/21/2023: Corn, silage

Field name: WW2 Crop: Corn, silage Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	357.77	61.16	755.41	5,770.02
Fresh water	1.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	365.89	61.16	755.41	5,770.02
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	262.51	45.26	233.09	1,222.04
Nutrient balance	103.37	15.89	522.31	4,547.98
Applied to removed ratio	1.39	1.35	3.24	4.72

Fresh water applied	
26,659,320.00	*gallons*
981.77	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
1,926,000.00	*gallons*
70.93	*acre-inches*
3.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Weeping Wall Manure

Sample and source description: 2022 Weeping Wall Manure

Sample date: 05/10/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	4,700.00	6,600.00							0.00
DL	0.01	0.01	0.01							1.00

Weeping Wall Manure

Sample and source description: Weeping Wall Manure

Sample date: 12/13/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,100.00	5,900.00	12,700.00	10,000.00	8,000.00	8,200.00	3,500.00	16,000.00		39.00
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 2022

Sample and source description: Lagoon Q4 2022

Sample date: 12/01/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	420.00	267.00	0.00	1.90	26.70	1,540.00								14,200.00	8,000
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	173.00	99.70	0.00	1.50	18.90	921.00								7,930.00	4,560
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 08/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	444.00	318.00	0.00	1.20	76.10	940.00								12,400.00	7,180
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,100.00	468.00	0.00	0.00	338.00	1,090.00								12,400.00	6,250
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

Canal

San Luis Canal

Sample description: San Luis Canal

Sample date: 08/03/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.10	0.00	0.10								213.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

HOME 1 - 10/27/2022: Oats, silage-soft dough

Home 1 Oat Silage

Sample and source description: Home 1 Oat Silage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	2,400.00	14,900.00		8.50
DL	0.01	0.01	0.01		1.00

HOME 1 - 05/21/2023: Corn, silage

HOME 1 - 05/21/2023: Corn, silage

Home 1 Corn Silage

Sample and source description: Home 1 Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,000.00	10,900.00		5.60
DL	0.01	0.01	0.01		1.00

HOME 2 - 10/27/2022: Oats, silage-soft dough

Home 2 Oat Silage

Sample and source description: Home 2 Oat Silage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	2,400.00	4,900.00		8.50
DL	0.01	0.01	0.01		1.00

NEMETHI 1 - 10/20/2022: Oats, silage-soft dough

Nemethi Ranch Field 1 Oat Silage

Sample and source description: Nemethi Ranch Field 1 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,300.00	23,000.00		13.80
DL	0.01	0.01	0.01		1.00

NEMETHI 1 - 05/25/2023: Corn, silage

Nemethi Ranch Field 1 Corn Silage

Sample and source description: Nemethi Ranch Field 1 Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,200.00	9,450.00		4.05
DL	0.01	0.01	0.01		1.00

NEMETHI 2 - 10/20/2022: Oats, silage-soft dough

Nemethi Ranch Field 2 Oat Silage

Sample and source description: Nemethi Ranch Field 2 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,300.00	23,000.00		13.80
DL	0.01	0.01	0.01		1.00

NEMETHI 3 - 10/20/2022: Oats, silage-soft dough

Nemethi Ranch Field 3 Oat Silage

Sample and source description: Nemethi Ranch Field 3 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,300.00	23,000.00		13.80
DL	0.01	0.01	0.01		1.00

NEMETHI 3 - 05/25/2023: Corn, silage

Nemethi Ranch Field 3 Corn Silage

Sample and source description: Nemethi Ranch Field 3 Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,200.00	9,450.00		4.05
DL	0.01	0.01	0.01		1.00

NEMETHI 4 - 10/20/2022: Oats, silage-soft dough

Nemethi Ranch Field 4 Oat Silage

Sample and source description: Nemethi Ranch Field 4 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,300.00	23,000.00		13.80
DL	0.01	0.01	0.01		1.00

NEMETHI 4 - 05/25/2023: Corn, silage

Nemethi Ranch Field 4 Corn Silage

Sample and source description: Nemethi Ranch Field 4 Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,200.00	9,450.00		4.05
DL	0.01	0.01	0.01		1.00

NEMETHI 5 - 10/20/2022: Oats, silage-soft dough

Nemethi Ranch Field 5 Oat Silage

Sample and source description: Nemethi Ranch Field 5 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	3,500.00	23,100.00		14.50
DL	0.01	0.01	0.01		1.00

NEMETHI 6 - 10/20/2022: Oats, silage-soft dough

Nemethi Ranch Field 6 Oat Silage

Sample and source description: Nemethi Ranch Field 6 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	3,500.00	23,100.00		14.50
DL	0.01	0.01	0.01		1.00

NEMETHI 6 - 05/25/2023: Corn, silage

Nemethi Ranch Field 6 Corn Silage

Sample and source description: Nemethi Ranch Field 6 Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,200.00	9,450.00		4.05
DL	0.01	0.01	0.01		1.00

NEMETHI 8/9 - 10/20/2022: Oats, silage-soft dough

Nemethi Ranch Field 8/9 Oat Silage

Sample and source description: Nemethi Ranch Field 8/9 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	3,500.00	23,100.00		14.50
DL	0.01	0.01	0.01		1.00

SANDERS - 10/27/2022: Ryegrass Silage

Sanders Field Ryegrass

Sample and source description: Sanders Field Ryegrass

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	3,300.00	32,500.00		13.90
DL	0.01	0.01	0.01		1.00

SANDERS - 05/21/2023: Corn, silage

Sanders Corn Silage

Sample and source description: Sanders Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	2,000.00	10,300.00		5.40
DL	0.01	0.01	0.01		1.00

WW1 - 10/27/2022: Ryegrass Silage

WW1 Field Ryegrass

Sample and source description: WW1 Field Ryegrass

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	3,500.00	34,200.00		16.60
DL	0.01	0.01	0.01		1.00

WW2 - 10/27/2022: Ryegrass Silage

WW2 Field Ryegrass

Sample and source description: WW2 Field Ryegrass

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	3,500.00	34,200.00		16.60
DL	0.01	0.01	0.01		1.00

WW2 - 05/21/2023: Corn, silage

WW2 Corn Silage

Sample and source description: WW2 Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	2,000.00	10,300.00		5.40
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

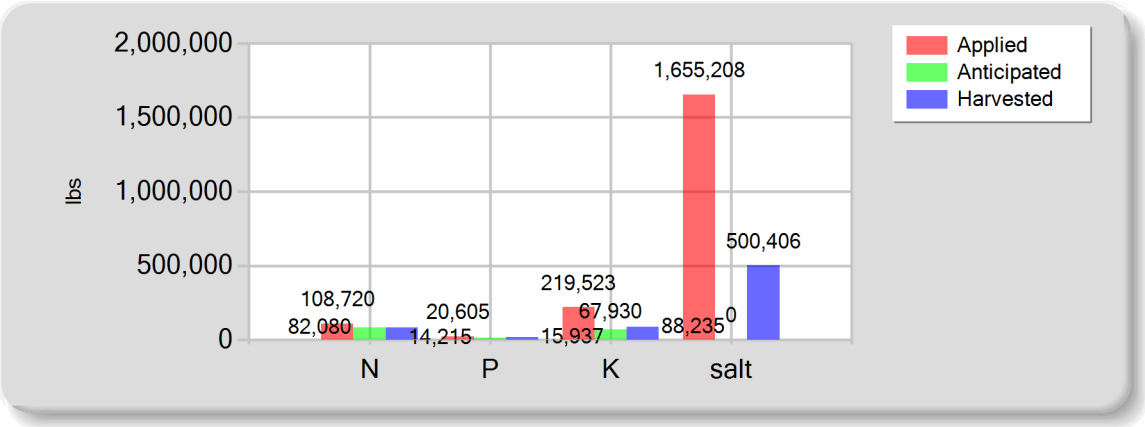
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

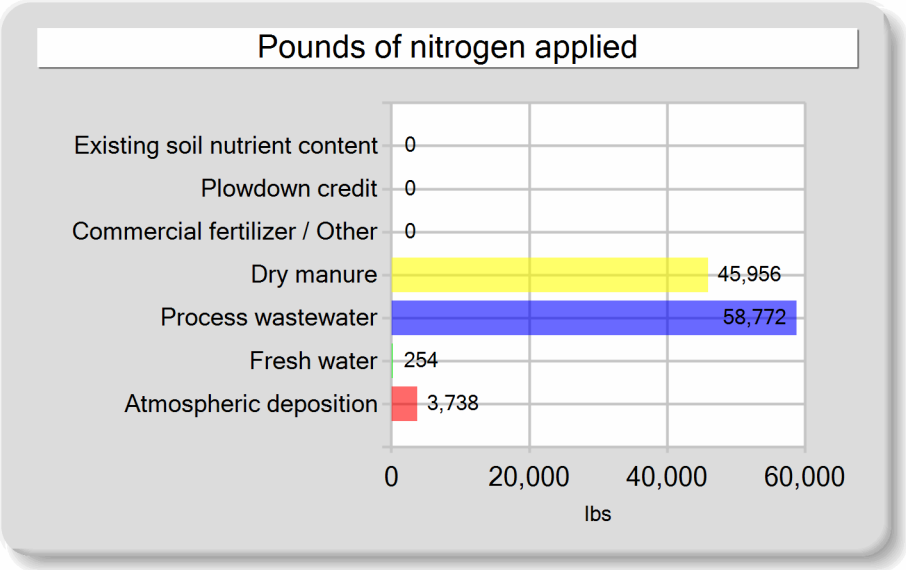
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

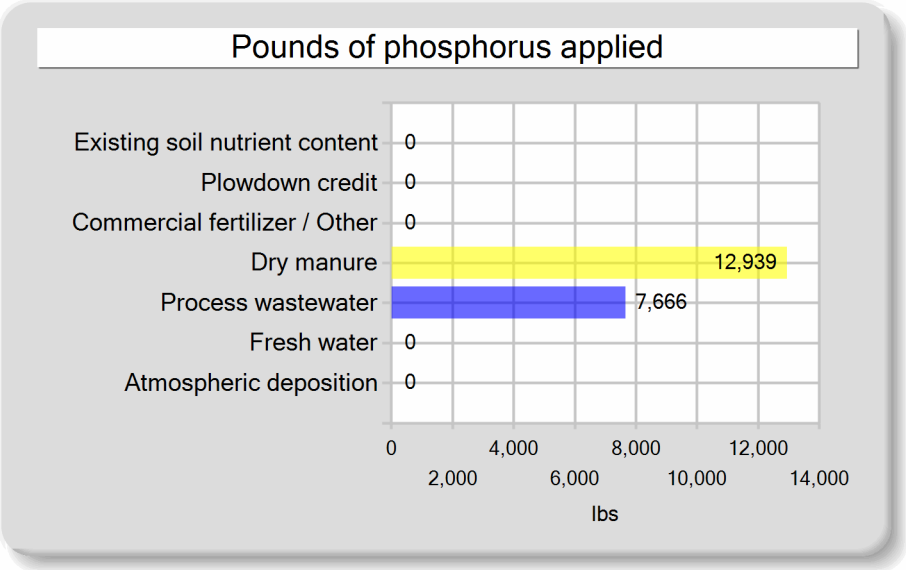
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	45,956.14	12,938.95	23,905.75	506,717.64
Process wastewater	58,772.48	7,666.43	195,617.73	1,148,490.64
Fresh water	253.67	0.00	0.00	0.00
Atmospheric deposition	3,738.00	0.00	0.00	0.00
Total nutrients applied	108,720.29	20,605.38	219,523.49	1,655,208.28
Anticipated crop nutrient removal	82,080.00	14,215.20	67,929.60	0.00
Actual crop nutrient removal	83,354.47	15,936.78	88,234.86	500,406.37
Nutrient balance	25,365.81	4,668.60	131,288.63	1,154,801.91
Applied to removed ratio	1.30	1.29	2.49	3.31

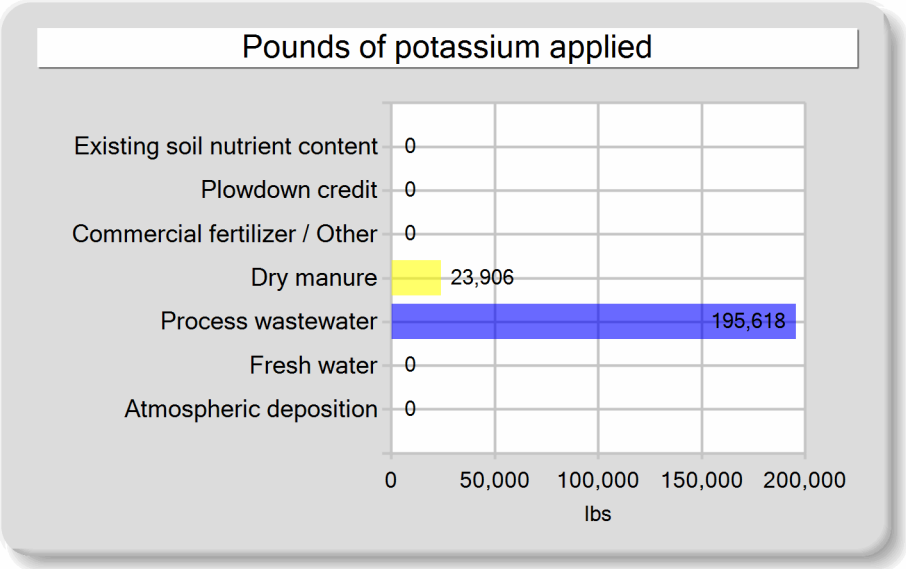
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

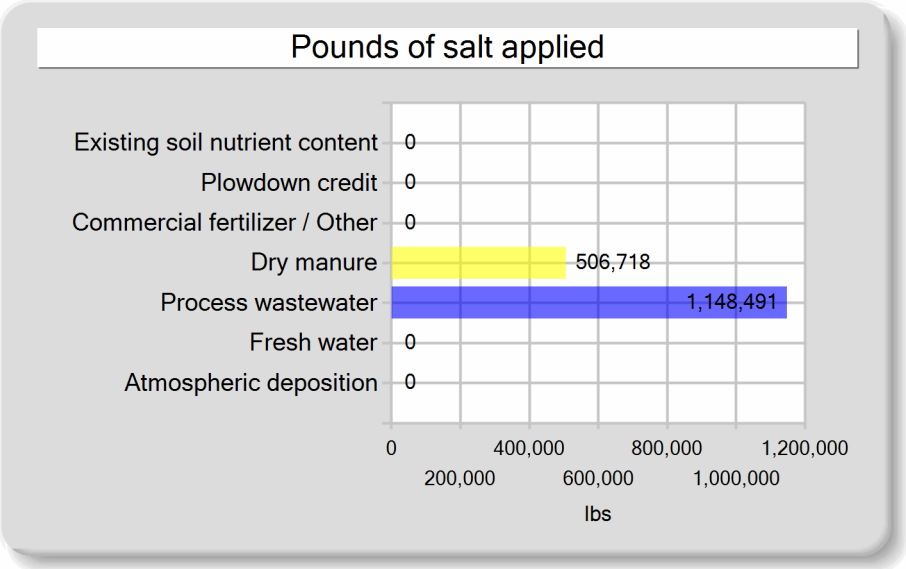


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Fields Home 2, Nemethi 2, Nemethi 5, Nemethi 8/9 and WW1 were summer fallowed

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

OUT OF COUNTRY-WILL SIGN WHEN RETURN

Anthony Goncalves

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Dennis & Kirsten Areias	Anthony Gonclaves
PRINT OR TYPE NAME	PRINT OR TYPE NAME
	Jul 16, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Den-K Holsteins
13315 Baker Rd
Los Banos CA 93635
00-0023502 08

E-mail: denk@elite.net
Copy To: cas.labs@yahoo.com

Lab No.: 23L0956
Sampled By: M.Pedroso
Requested By: Kirsten Areias
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project: Den-K Heifers
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW-MB	12/13/23	15:50	2340	2.34	0.6	ND		7.7

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Den-K Holsteins
13315 Baker Rd
Los Banos CA 93635
00-0023502 08

E-mail: denk@elite.net
Copy To: cas.labs@yahoo.com

Lab No.: 23L0956
Sampled By: M.Pedroso
Requested By: Kirsten Areias
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project: Den-K Heifers
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW-MB	12/13/23	15:50	**2340**	**2.34**	0.6	1.64	0.14

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Den-K Holsteins
13315 Baker Rd
Los Banos, CA 93635

Account# 00-0023502
Account Manager: Ben Nydam
Submitted By: Kirsten Areias
Ranch: Den-K Heifers

Received: 12/14/2023 14:20
Reported: 12/21/2023 09:47

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0956-01	DW-MB	Ag Water	M.Pedroso		12/13/2023 15:50

Default Cooler Temperature on Receipt °C: 5.4
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Sample Results

Sample: DW-MB
23L0956-01 (Water)

Sampled: 12/13/2023 15:50
Sampled By: M.Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.34**	mmhos/cm	0.01	1		12/18/23 15:42	SM 2510 B		BEL0680
Electrical Conductivity umhos	**2340**	umhos/cm	10.0	1		12/18/23 15:42	SM 2510 B		BEL0680
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 15:50	Field		BEL0624
Nitrate Nitrogen as NO3N	**0.6**	mg/L	0.1	1	10	12/15/23 17:58	EPA 300.0		BEL0597
Temperature	**25.0**	units	0.0	1		12/18/23 15:42	SM 4500-H+	H	BEL0680
pH	**7.7**	units	1.0	1		12/18/23 15:42	SM 4500-H+	H	BEL0680

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0597										
Blank (BEL0597-BLK1)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0597-BLK2)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0597-BLK3)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0597-BLK4)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0597-BLK5)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0597-BS1)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.6	90-110		
LCS (BEL0597-BS2)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
LCS (BEL0597-BS3)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEL0597-BS4)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000		107	90-110		
Duplicate (BEL0597-DUP1)		**Source: 23L0943-02**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	4.4		0.1	mg/L		4.4			1.52	10
Duplicate (BEL0597-DUP2)		**Source: 23L0943-13**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	2.3		0.1	mg/L		2.2			1.96	10
Duplicate (BEL0597-DUP3)		**Source: 23L0954-06**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	1.6		0.1	mg/L		1.6			1.76	10
Duplicate (BEL0597-DUP4)		**Source: 23L0957-03**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	3.5		0.1	mg/L		3.5			1.20	10
Matrix Spike (BEL0597-MS1)		**Source: 23L0943-02**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.2		0.1	mg/L	5.000	4.4	116	90-110		
Matrix Spike (BEL0597-MS2)		**Source: 23L0943-13**			Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	7.3		0.1	mg/L	5.000	2.2	102	90-110		

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Ranch: Den-K Heifers

Received: 12/14/2023 14:20
Reported: 12/21/2023 09:47

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0597 (Continued)										
Matrix Spike (BEL0597-MS3)			**Source: 23L0954-06**		Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	6.4		0.1	mg/L	5.000	1.6	96.7	90-110		
Matrix Spike (BEL0597-MS4)			**Source: 23L0957-03**		Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	8.7		0.1	mg/L	5.000	3.5	104	90-110		
Reference (BEL0597-SRM1)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.2	90-110		
Reference (BEL0597-SRM2)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.0			mg/L	10.00		100	90-110		
Reference (BEL0597-SRM3)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0597-SRM4)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0597-SRM5)					Prepared: 12/14/2023 Analyzed: 12/15/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0680										
Blank (BEL0680-BLK1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.6		1.0	units						
Blank (BEL0680-BLK2)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.7		1.0	units						
Blank (BEL0680-BLK3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0680-DUP1)		**Source: 23L0955-04**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	0.25		0.01	mmhos/cm		0.25			0.848	10
pH	8.3		1.0	units		8.2			0.363	10
Electrical Conductivity umhos	249		10.0	umhos/cm		246			0.848	10
Duplicate (BEL0680-DUP2)		**Source: 23L0958-02**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	0.25		0.01	mmhos/cm		0.25			1.07	10
Electrical Conductivity umhos	253		10.0	umhos/cm		250			1.07	10
pH	8.1		1.0	units		8.1			0.123	10
Reference (BEL0680-SRM1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	431			umhos/cm	426.0		101	90-110		
Reference (BEL0680-SRM2)					Prepared & Analyzed: 12/18/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0680-SRM3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0680-SRM4)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0680-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0680 (Continued)										
Reference (BEL0680-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0680-SRM6)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0680-SRM7)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0680-SRM8)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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12/14/23 14:20 23L0956

WATER WORK REQUEST

Bill To: Acct No. 23502 Cons. 8

Purchase Order No. Results Needed By

Client: Den-K Holsteins
Address: 13315 Baker Rd
City, State, Zip: Los Banos, CA 93635
Phone: Fax:
Cell/Email:

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Kirsten Areias (209) 826-6242

Ranch: Den-K Heifers

Date sampled: 12/13/23

Sampled by: M Pedroso

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6126 • 800 228-9896 • Fax 559 268-8174

No. of Samples 1 No. Bottles 1

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Analysis and Bottles Required: *(Please Indicate Analysis)*

(Y) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-MB	Sampled From:	12/13	3:50	0	5.4
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M Pedroso	CAS Inc	12/13/23 @ 3:50	12/14/23 12:8
Second	JLF	PLV	12-14-23 12:00	12-14-23 2:15
Third				
Fourth		DCL	12/14 14:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Shipping
$ In
$ Out

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid ___ Rec By ___ Check No. ___ Date ___

Signature

Sample received in cooler with ice?
[ ] Yes [ ] No

ctl:update 2020

Shipping Information: Shipped In □ Picked-Up ☒ Walk In □ DLI Sampler □ Other □ _____

Container: Ice Chest ☒ Box □ None □ **Refrigerant:** Wet Ice ☒ Blue Ice □ None □

Samples Preserved with HNO_3 or H_2SO_4 were: □ Received Preserved □ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/14/23 14:20 23L0956

SEND OUT